ALEX ARIZA HERRERA

AI Engineer | LLM & RAG Systems | Full-Stack Development

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Professional Summary

AI Engineer focused on building LLM and RAG systems for regulated domains—healthcare, public sector, and internal knowledge—where a wrong answer has real cost, not just bad UX. Built a dental EHR on AWS with HL7 FHIR RDA and MinSalud IHCE integration; enterprise RAG copilot on FastAPI/pgvector with page-level citations and a live demo; local document engine with hybrid retrieval and an evaluation harness; and a multi-agent SGR/MGA pipeline with versioned normative canon and mandatory human reviews. Full-stack from ingestion to deployment.

Independent Experience

Independent Software Engineer 2025 – Present

- Design and deliver full-stack products with LLM/RAG integration for own projects and clients in healthcare, public sector, and document knowledge.
- Full cycle—architecture, APIs, frontend, CI/CD, observability—with public case studies at alexariza.dev; focus on traceability, evidence grounding, and human review.

Selected Projects

AI Clinic Operations Copilot — Compliance-First Dental EHR (Colombia)

View Project

EHR platform for Colombian dental clinics: regulatory compliance, legal traceability, and MinSalud interoperability.

- **Impact:** Reduces regulatory risk against fines up to **2,000 SMLMV**, activity suspension, and loss of IHCE interoperability under Resolution 1888/2025 for paper or non-compliant clinics.
- Designed append-only clinical folios: signed records cannot be edited or deleted; rectifications are new linked records that preserve legal traceability and chronological integrity.
- Implemented consent, certified digital signature (PKCS#7/CAdES), RBAC, immutable audit trails, and privacy-by-design aligned with Law 2015/2020, Law 1581/2012, and Colombian health resolutions.
- Implemented HL7 FHIR RDA submission to IHCE (MinSalud) with VIDA tracking, transactional outbox for reliable government integration, and in-clinic patient record delivery with witness audit.
- End-to-end deployment on AWS (EC2, RDS PostgreSQL, S3, ECR, GitHub Actions) with OpenAPI as the contract source of truth between FastAPI and Next.js.

Stack: Python 3.12, FastAPI, SQLAlchemy Async, Alembic, PostgreSQL 16, Redis, Celery, Next.js 16, TypeScript, Docker, OpenAPI, HL7 FHIR, AWS

AI System for SGR/MGA Project Formulation — Multi-Agent Planning Assistant

View Project

SGR/MGA pre-investment formulation from incomplete municipal inputs—traceable sources, explicit gaps; scope stops before approval or financing decisions.

- **Impact:** Enables small and medium municipalities—without installed technical capacity—to turn territorial needs into defensible, traceable pre-investment profiles to compete for royalty funding.
- Designed a 15-step deterministic pipeline that structures problem trees, objectives, alternatives, MGA logic, indicators, risks, and gap reviews from incomplete inputs.
- Evidence-first workflows: every assumption, requirement, and recommendation traceable to a source document, normative canon rule, or explicitly marked gap—no invented fields.
- Orchestrated specialized subagents with bounded context via OpenCode; read-only canon in `knowledge/` and

versioned artifacts in auditable Markdown.

- Mandatory human review before methodological closure; active consultancy pilots—does not promise approval or disbursement.

Stack: OpenCode, Agent Orchestration, Structured Prompting, Knowledge Modeling, Document Analysis, Markdown/Git Artifacts, Human-in-the-Loop

AI Internal Knowledge Copilot — Enterprise RAG Application

View Project | Live Demo

Knowledge assistant over internal PDFs: ingestion, vector retrieval, streaming answers, and page-level source citations.

- **Impact:** McKinsey estimates 1.8 h/day searching for internal information; recovering 20 % in a 100-person org equals ∼**9,360 hours/year**.
- Built JWT auth, per-user document ownership, PDF ingestion with pgvector retrieval, SSE streaming chat, analytics events, and Docker Compose deployment.
- Extract PDFs to structured Markdown before chunking so headings, page markers, and section hierarchy carry into retrieval context.
- Embeddings via VoyageAI and generation via OpenAI; typical latency 1.5–2 s and ∼$0.0003 per query at 1,000 queries/day scale.
- Separated auth, documents, chat, and analytics into router → service → repository modules on FastAPI; no-answer policy when corpus lacks evidence.

Stack: TypeScript, Next.js, React, FastAPI, PostgreSQL, pgvector, VoyageAI, OpenAI, Docker, SSE Streaming

Local RAG Knowledge Copilot — On-Machine Document Q&A

View Project

Local web app for confidential contracts, policies, and manuals: cited Q&A without sending document content to the cloud by default.

- **Impact:** Legal, compliance, and operations teams query policies with page- and section-level citations without exposing document bytes to external APIs.
- FastAPI backend and React UI: PDF upload, document library, natural-language query, and retrieved chunks shown with each answer.
- Hybrid semantic + lexical retrieval with RRF fusion and heading-aware ranking for Spanish policy corpora; deduplication via SHA-256 content hash.
- Explicit no-evidence response when corpus does not support the question; golden-set eval harness, 12+ pytest modules

Stack: Python 3.12, FastAPI, Docling, ChromaDB, SQLite FTS5, sentence-transformers, llama.cpp, React, TypeScript, pytest

Education

Universidad de los Andes, Electronic Engineering Degree

The University of Texas at Austin, Certificate in Full Stack Software Development: Building Scalable Cloud Applications

Skills

Languages: Python, TypeScript, SQL | **AI & LLM:** RAG, Hybrid Retrieval, Vector Search, Embeddings, Agent Orchestration, Prompt Engineering, Local LLM Inference (`llama.cpp`), Eval Harnesses
Backend: FastAPI, PostgreSQL, pgvector, SQLAlchemy, Redis, Celery, REST APIs, JWT Auth, RBAC, OpenAPI |
Frontend: React, Next.js, TypeScript, Tailwind CSS, TanStack Query
Infrastructure & Quality: Docker, AWS (EC2, RDS, S3, ECR), GitHub Actions, CI/CD, pytest